Yang Zhang

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Education

University of California San Diego — Dec. 2025 – Present
M.S. in Computer Engineering; GPA 3.94/4.0 — La Jolla, CA

University of Electronic Science and Technology of China (UESTC) — Sept. 2021 – Jun. 2025
B.Eng. in Communication Engineering; GPA 3.8/4.0 (top 10%) — Chengdu, China

- Coursework: Artificial Intelligence and Machine Learning, Stochastic Signal Analysis, Communication Networks, Embedded Processors.
- GRE 330; IELTS 8.0.

Research Interests

Natural language processing, large language models, and reinforcement learning: RL post-training with verifiable rewards (GRPO/RLVR) for LLM reasoning; synthetic data and self-evolving curricula; the token and compute budgets behind post-training and inference; contamination-free evaluation and benchmark construction; interpretability of the computations learned by language models.

Publications and Preprints

* denotes equal contribution.

[1] **What Do Rollout Tokens Buy? Hidden Costs and Budgeted Accuracy in RLVR**
Yang Zhang and co-authors (author list to be added).
Under review.
Summary: In RLVR on competition mathematics, more rollout tokens do not buy more correct answers; we propose judging training choices by the tokens they add on unchanged problems and by the answers returned under a fixed token budget.

[2] **Latent Assembly: Causal Discovery and Cross-Task Linking of Executable Operations in Language Models**
Yang Zhang and co-authors (author list to be added).
Under review.
Summary: A causal framework that decompiles language-model computation into typed states, executable operations, and the neural interfaces between them, and links independently discovered operations into new programs.

[3] **DeepImagine: Clinical Trial Outcome Prediction via Stepwise Local Counterfactual Imaginations** [arXiv][code]
Under review; arXiv preprint, 2026.
Summary: Predicts clinical-trial outcomes by imagining, one factor at a time, how a historical trial's observed result would change toward the target trial; beats direct prediction across off-the-shelf LLMs.

[4] **Question Begets Question: Self-Evolving Curriculum for Reinforcement Fine-Tuning on Competition Mathematics** [arXiv]
Longtian Bao*, Jianyou Wang*, **Yang Zhang***, Youze Zheng*, Ramamohan Paturi.
Under review; arXiv preprint, 2026.
Summary: A self-evolving curriculum of teacher-generated problem variants for RL-only fine-tuning; lifts Qwen2.5-Math-7B pass@1 on AIME from 5.6% to 16.5% under a fixed data budget, breaking the plateau of static augmentation.

[5] **CT Open: An Open-Access, Uncontaminated, Live Platform for the Open Challenge of Clinical Trial Outcome Prediction** [arXiv][website][code]
Jianyou Wang*, Youze Zheng*, Longtian Bao*, Hanyuan Zhang*, Qirui Zheng, Yuhan Chen, **Yang Zhang**, Matthew Feng, Maxim Khan, Aditya K. Sehgal, Christopher D. Rosin, Ramamohan Paturi, Umber Dube, Leon Bergen.
COLM 2026.
Summary: A live, contamination-free benchmark for prospective clinical-trial outcome forecasting, backed by an automated, expert-validated decontamination pipeline; releases a training set and two time-stamped test sets.

[6] **Improved RT-DETR Based on MobileNetV4 for Vehicle Detection** [IEEE Xplore]
Yang Zhang.
ICAACE 2025 (2025 8th International Conference on Advanced Algorithms and Control Engineering), IEEE.
Summary: A MobileNetV4 backbone for RT-DETR that keeps detection accuracy with 65% fewer parameters and 75% less computation than RT-DETR-L.

Research Experience

Laboratory for Emerging Intelligence, UC San Diego 2026 – Present
Research Assistant; advisor: Prof. Ramamohan Paturi La Jolla, CA

- **RL post-training for reasoning (Question Begets Question, co-first author):** built a self-evolving curriculum for RL fine-tuning of Qwen2.5-Math-7B on AIME using only problem statements and final answers; teacher-generated variants seeded from problems the current checkpoint can mostly solve lift pass@1 from 5.6% to 16.5% under a fixed data budget (static augmentation plateaus at 12.5–14.5%).
- **Cost-aware RLVR (What Do Rollout Tokens Buy?):** measured the hidden token costs of RLVR training choices on SmolLM3-3B and Qwen3.5-9B: a harder training mix doubles response length on unchanged problems, while halving the training response limit cuts compute by 28% and raises majority-vote accuracy from 36.1% to 43.5% under a fixed evaluation budget.
- **Contamination-free evaluation (CT Open, COLM 2026):** contributed to a live clinical-trial outcome prediction platform and its automated decontamination pipeline (iterative LLM web search, temporal filters, expert validation) that timestamps the earliest public mention of each trial outcome; released a training set and two time-stamped test benchmarks.
- **Counterfactual reasoning (DeepImagine):** contributed to a stepwise local counterfactual imagination framework for clinical-trial outcome prediction and to training a family of specialized per-factor language models on natural counterfactuals with synthetic reasoning traces.
- **Mechanistic interpretability (Latent Assembly):** contributed to a causal framework that recovers typed program states and executable operations from unlabeled activations in Qwen, Gemma, and Llama models and links independently discovered operations into a three-operation chain reaching 84% accuracy with no target-task fitting.

Institute of Computing Technology, Chinese Academy of Sciences Jul. 2024 – Aug. 2024
Research Assistant, State Key Laboratory of Processors Beijing, China

- Developed and optimized image signal processing algorithms (noise reduction, sharpening, color correction) in Python, MATLAB, and C for embedded systems built around CMOS image sensors.

Institute of Computing Technology, Chinese Academy of Sciences Aug. 2023 – Nov. 2023
Research Assistant, Center for Advanced Computer Systems Beijing, China

- Improved the Maximum Information Coefficient (MIC) algorithm for identifying and visualizing correlations across datasets, and optimized the data-processing workflow for accuracy and efficiency.

Selected Projects

Vehicle Detection with Improved RT-DETR Based on MobileNetV4 Apr. 2024 – Sept. 2024

- Replaced the RT-DETR backbone with a lightweight MobileNetV4 for real-time vehicle detection on edge devices: 11M parameters and 38 GFLOPs, 65%/75% fewer parameters/GFLOPs than RT-DETR-L and 45%/54% fewer than RT-DETR-R18, at comparable accuracy on a COCO-format vehicle dataset. Published at ICAACE 2025 [6].

Honors and Awards

- UESTC Merit Scholarship for Exceptional Student (top 10%), Oct. 2024.
- UESTC Merit Scholarship (top 30%), Oct. 2022 and Oct. 2023.
- Hainan Free Trade Port Merit Scholarship (top 10%), Jan. 2022.

Service

- Student Volunteer (session monitor), ACL 2026 (64th Annual Meeting of the Association for Computational Linguistics), San Diego, CA, Jul. 2026.

Technical Skills

- **Languages:** Python, C, MATLAB, Bash, LaTeX.
- **ML/LLM frameworks and tools:** PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, Git.
- **LLM methods:** RL with verifiable rewards (GRPO/RLVR), supervised fine-tuning, synthetic data and curriculum learning, pass@k and majority-vote evaluation, benchmark construction and decontamination, LLM-as-judge, interpretability.
- **Computer vision:** real-time object detection (RT-DETR, MobileNetV4); image signal processing.